Social Media & Network Analytics

Worksheet

Do Polarized Emotional Communities Form Around AI Job Replacement Discussions on YouTube, and Who Are the Influential Users Shaping These Communities?

File Name:	Worksheet_S4186963_S4190295_PG_Group_39
Course:	COSC2671 — Social Media & Network Analytics
Level:	Postgraduate (PG)
Group:	Group 39
Members:	**Chantelle Isabelle Dmonte (S4186963)** **Utkarsh Nautiyal (S4190295)**

SECTION 1 — TEAM PROJECT PLAN

Research Question: Do polarised emotional communities form around AI job replacement discussions on YouTube and Bluesky?

Platforms: YouTube (comments) & Bluesky (posts & replies)

Week	Task / Milestone	Responsible	Status
Wk 9	**6 -7 May 2026 — Research question finalisation & data collection setup**		
	Agree on research question & platform focus (YouTube + Bluesky)	Both	Completed
	Configure YouTube Data API v3 & Bluesky AT Protocol credentials	Utkarsh	Completed
	Design data collection strategy (keywords, date range, volume targets)	Chantelle	Completed
	Initial literature review on polarisation metrics	Chantelle	Completed
Wk 9	**8-10 May 2026 — Data collection & preprocessing**		
	Collect YouTube comments & Bluesky posts (target ≥1,000 each)	Utkarsh	Completed
	Text preprocessing pipeline (tokenisation, stopword removal, lemmatisation)	Utkarsh	Completed
	VADER sentiment analysis on both datasets	Chantelle	Completed
	Dataset overview statistics & initial EDA	Utkarsh	Completed
Wk 10	**11-17 May 2026 — Network construction & NLP analysis**		
	Build reply-graph networks (YouTube & Bluesky)	Chantelle	Completed
	Compute centrality measures (PageRank, betweenness, in-degree)	Chantelle	Completed
	LDA topic modelling with perplexity validation	Utkarsh	Completed
	Community detection via Louvain algorithm	Both	Completed
Wk 11	**18-24 May 2026 — Analysis, visualisation & report writing**		
	Community sentiment analysis & polarisation index	Chantelle	Completed
	Statistical tests (Kruskal-Wallis, Mann-Whitney)	Chantelle	Completed
	Network visualisations (centrality scatter, community bar charts)	Utkarsh	Completed
	Cross-platform comparison heatmap	Chantelle	Completed
	Draft report — Sections 1–3 (Introduction, Data, Methods)	Both	Completed
Wk 12	**25 May 2026 — Final report, submission & worksheet**		
	Complete Sections 4–5 (Results & Discussion)	Chantelle	Completed
	Finalise worksheet (timesheets + self-reflections)	Both	Completed
	Code cleanup, GitHub submission	Both	Completed

Week	Task / Milestone	Responsible	Status
	Final submission	Both	Completed
	Slide creation for presentation	Both	In progress

SECTION 2 — WEEKLY TIMESHEETS

Timesheet — CHANTELLE ISABELLE DMONTE (S4186963)

Week	Dates	Tasks Completed	Mon–Wed (hrs)	Thu–Fri (hrs)	Total Hrs
Week 9	6–10 May 2026	Research question finalisation API credential setup Data collection strategy design Literature review on polarisation	12	5	17
Week 10	11–17 May 2026	Data collection (YouTube + Bluesky) Text preprocessing pipeline VADER sentiment analysis EDA & dataset overview	10	3	13
Week 11	18–24 May 2026	Reply-graph network construction Centrality measures computation LDA topic modelling Community detection (Louvain)	15	5	20
Week 12	25 May–30 May 2026	Community sentiment analysis Statistical tests Visualisations & charts Draft report sections Final report completion Peer review Worksheet finalisation Submission	10	5	15
		TOTAL HOURS			65

Timesheet — UTKARSH NAUTIYAL (S4190295)

Week	Dates	Tasks Completed	Mon–Wed (hrs)	Thu–Fri (hrs)	Total Hrs
Week 9 6–10 May 2026		Research question finalisation API credential setup Data collection strategy design Literature review on polarisation	10	5	15
Week 10 11–17 May 2026		Data collection (YouTube + Bluesky) Text preprocessing pipeline VADER sentiment analysis EDA & dataset overview	8	4	12
Week 11 18–24 May 2026		Reply-graph network construction Centrality measures computation LDA topic modelling Community detection (Louvain)	10	5	15
Week 12 25 May–30 May 2026		Community sentiment analysis Statistical tests Visualisations & charts Draft report sections Final report completion Peer review Worksheet finalisation Submission	10	5	15
TOTAL HOURS					57

SECTION 3 — INDIVIDUAL SELF-REFLECTION

Self-Reflection — CHANTELLE ISABELLE DMONTE (S4186963)

Working on this project gave me a meaningful opportunity to apply social media analytics techniques to a real-world research question about AI discourse and community polarisation. My primary contributions spanned data collection pipeline design, preprocessing, VADER sentiment analysis, and LDA topic modelling with perplexity-based validation — all implemented in Python using a checkpoint system to preserve YouTube API quota across runs.

One of the key technical challenges I encountered was the ATS-clean data ingestion issue with Bluesky's reply_to_handle field: the raw AT Protocol URI format required a custom extract_reply_target() function before a valid reply graph could be constructed. Debugging this iteratively — printing diagnostics, inspecting dtypes, and patching the graph builder call — reinforced my appreciation for defensive data engineering practices.

Collaborating with Utkarsh helped me improve my approach to task decomposition and asynchronous code review. In hindsight, I would invest more time upfront in agreeing on a shared data schema before each of us began independent collection modules, which would have reduced integration friction in Week 11.

Overall, this project deepened my understanding of network-based approaches to studying online discourse, and strengthened my practical skills in graph analytics and NLP pipelines — areas directly relevant to my Master of AI studies and future graduate roles in data/AI engineering.

Self-Reflection — UTKARSH NAUTIYAL (S4190295)

This project provided an excellent opportunity to bridge theoretical knowledge of network analysis with hands-on implementation across two distinct social media platforms. My primary contributions included network construction (building and validating both the YouTube and Bluesky reply graphs), centrality computation, Louvain community detection, and the cross-platform comparison visualisations.

A significant challenge was ensuring methodological consistency between the two platforms despite their structural differences YouTube's nested comment threads versus Bluesky's decentralised AT Protocol reply chains. Adapting the graph builder to handle both formats without losing directed edge information required careful review of each platform's data model.

Working with Chantelle was productive and well-structured; her systematic approach to preprocessing and sentiment analysis made the downstream network tasks more reliable. One area for improvement in future projects would be establishing shared integration tests early, so that each module's outputs are validated against expected schemas before being passed to the next stage.

This experience has consolidated my skills in NetworkX, Python-Louvain, and statistical testing (Kruskal-Wallis), and has given me a richer understanding of how online communities organise around emotionally charged topics insight I look forward to building on in my research and professional career.